



FABRIC V8.2.0 RELEASE NOTES

These Release Notes describe the new features in Fabric release V8.2.0 and list bugs that have been fixed since the V8.1.0 release.

Certification of this Fabric release is based on:

- Cassandra version 4.1.3
- SQLite version 3.46.1
- OpenJDK Runtime Environment 21.0.3
- Confluent Kafka version 7.6
- Neo4j 5.25.1 - enterprise
- Elasticsearch - 8.5.3
- AWS OpenSearch – 2.17
- PostgreSQL 15.4

MAIN FEATURES AND IMPROVEMENTS

1. Fabric Integration with GenAI

Fabric has been strengthened with a list of new capabilities that are based on LLM GenAI integration. This integration utilizes the new available GenAI connectors in the k2exchange. The new capabilities elevate various Fabric modules during the design time (Catalog, Web Studio and Broadway flows creation) as well as at run time (Broadway actors). These capabilities are detailed below:

- **LLM-based Catalog profiling** – a new product plugin, which performs profiling of each column's metadata and data using an LLM model via an LLM AI project interface (which requires installation of the relevant extension). The plugin response depends on the user prompt and other input parameters. The product configuration includes 2 instances of this plugin (both are inactive by default):
 - **LLM Data Profiling** - classification of columns with sensitive / PII data, using the column name and values.
 - **LLM Description** - creation of each column's short description, based on the column name and values.
 - The same plugin can be used to accommodate additional use cases via updating the user prompt and other plugin's input parameters. One of these use cases is profiling using a column's property – e.g. description. The plugin can be useful when columns don't have meaningful names, whereas the property (e.g. description) sheds light about the column's meaning. More information can be found in the Support article.

https://support.k2view.com/Academy/articles/39_fabric_catalog/plugins/02_LLM_data_profiling.html



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- **AI Assistant for SQL statements** - a newly built-in AI Assistant within the Query Builder editors that enables an effortless SQL statements creation by simply describing the desired query in a natural language. The user can also use the AI Assistant to explain existing SQL statements by translating them into plain, easy-to-understand natural language.
- **AI-generated Broadway flow info** can be created from the Actions menu of a Broadway flow. It includes the flow overview and a detailed explanation about each stage.
- **Broadway Co-pilot** is an AI-based Broadway capability, which searches for required actors using a natural language rather than standard key words. When multiple actors are suggested by Co-pilot, they can be added to a flow at once.

https://support.k2view.com/Academy/articles/19_Broadway/03_broadway_actor.html

2. Fabric Catalog

Fabric's Catalog has been enhanced to perform discovery based on user-defined rules. The Catalog's UI facilitates the creation of rules that allow overriding the product baseline and can consequently impact the process on various hierarchy levels: data platform, schema or even dataset.

Examples for such rules are:

- Deactivate (or activate) selected plugins for a specified schema.
- Define a sample size that differs from the baseline - for a specified schema or a dataset of a selected data platform.
- Set up a crawler filter (a new way to define *include* and *exclude* lists).

Moreover, the Catalog introduces several enhancements: Catalog-based sequences, setup of seed random values, centralized Catalog for multiple Fabric instances and more. These capabilities are detailed below:

- **Discovery Pipeline configuration screen** enables updating the Discovery Job configuration (including baseline override and new rules' creation) via a UI rather than by editing a JSON file. Using this screen, one can override the default Baseline rule, create new rules and add new plugins to the pipeline.
 - A baseline configuration override can be applicable for any data platform. The job executes a combination of baseline and user-defined rules.
 - The product level `discovery.plugins` file has been transformed to a new format. The project level file is now called **discoveryOverrides.plugins** and it replaces the previous project level file. Due to these format changes, the projects with a Catalog upgrade to V8.2 should follow the [Fabric Upgrade procedure to 8.2](https://support.k2view.com/Academy/articles/39_fabric_catalog/13_discovery_pipeline_settings.html).

https://support.k2view.com/Academy/articles/39_fabric_catalog/13_discovery_pipeline_settings.html

- **Catalog-based sequence** – a new **Sequences** tab (located under the Catalog Settings) that enables setting up the sequences.



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- The Catalog masking actors have been enhanced to generate the sequences in population flows using on the Catalog-based sequences setup.
- Note that currently the Catalog does not identify the sequence fields automatically, therefore fields should be manually marked as sequences (via the Edit Catalog action).

https://support.k2view.com/Academy/articles/39_fabric_catalog/10_catalog_settings.html

- **Catalog settings for seed random values** – Data consistency using seed has been introduced in this version. It ensures referential integrity without storing mappings in the caching table. By leveraging the **Java Random** method with a **seed** (populated from the caching key), it delivers better performance compared to regular masking.
 - To utilize this mechanism, set the **Consistency Mode** of the Masking Generator to **Consistent using seed** value using the Catalog's PII & Masking tab (this value is only available for Generators that support seed). The Consistency Mode replaces Consistent and Unique checkboxes in the tab.

https://support.k2view.com/Academy/articles/39_fabric_catalog/10_catalog_settings.html

https://support.k2view.com/Academy/articles/26_fabric_security/06_data_masking.html

- **Enhanced Catalog masking and sequence generation** – sending the entire record for the data generator in addition to the original value. This enables intelligent masking and sequence generation where the masked value of one field can be determined based on other fields within the same record.
- **Central Catalog for multiple clusters** – it is now possible to configure a centralized Catalog for multiple Fabric instances. This is useful when, for example, several users need to work on the same project in parallel.
 - Working in separate Fabric instances (e.g., spaces), the users can define different Catalog settings and run Discovery separately one each Fabric instance. At the end, the Catalog artifacts can be combined into one.
 - Additionally, users may be connected to the Catalog in a read-only mode, enabling then to view the Catalog tree but not update it by either running Discovery or doing manual overrides. The read-only permissions affect only the Neo4j, while these users can still update the settings or create an artifact.

https://support.k2view.com/Academy/articles/39_fabric_catalog/98_centralized_catalog_setup.html

- **Data Quality Metrics** – a new product plugin, which scans the data snapshot while calculating several data quality metrics (distinct values count and a range of values) and creating them as field's properties in the Catalog. The null percentage calculation has been combined with the new metrics and is now a part of this plugin.

https://support.k2view.com/Academy/articles/39_fabric_catalog/plugins/04_source_data_metrics.html

- **Fabric memory handling** has been improved with the ability to balance the Fabric memory either when running Discovery on a data platform with multiple schemas or when multiple Discovery jobs are running in parallel on the same Neo4j.



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- A new configuration parameter #DATA_SNAP_WRITE_MEMORY_CAP_MB=4096 defines the maximum amount of the Fabric memory allocated for the Data Snapshot process. When working with large data sources, it is recommended to increase this setting, assuming the system has sufficient resources for such increase.
- Note that the DATA_SNAPSHOT_WRITE_BATCH_SIZE_MB parameter has been removed from the Fabric configuration.

https://support.k2view.com/Academy/articles/39_fabric_catalog/21_advanced_settings.html

- **Running Discovery on Fabric and its entities** – the Discovery job is now performed on all Fabric entities including Graphit and populations. This feature is the basis for the Data Lineage that to be included in a future version, and it will then present the data flow from any source to Fabric.

3. Broadway

New Actors

- **InnerFlowSession** actor - opens a new Fabric session and enables running an attached inner flow asynchronously, either locally (default) or on a random node.

4. Web Studio

- The user code in the Web Studio is now compiled with OpenJDK 21.0.3.
- **Filter the interface explorer by a property** – tables in the DB Explorer can now be filtered, based on the Catalog's Dataset properties. Table filtering by a property helps to create a Catalog-based Logical Unit, with a smaller number of tables.
 - The filter is applicable for the interfaces after running Discovery.
 - A **Fiter** icon has been added to the schema level in the Interface Explorer. When clicked, the filter is set for usage and its color changes to orange.

https://support.k2view.com/Academy/articles/04_fabric_studio/25_web_data_explorer.html

- The **field and table descriptions** are now propagated from the Catalog to the Web Studio and are captured in the Logical Unit during the LU Schema creation.
- **CSV Editor** - a new editor that introduces several previously unavailable capabilities, including an automatic recognition of user-made changes; adding rows and columns; reordering rows and columns via drag-and-drop; sorting of columns; copying and pasting table cells, rows, or columns.

5. Miscellaneous

- **StreamSync** – a solution that enables a proactive synchronization of Fabric with source systems by processing the changes only, without the need to re-synchronize the entire instance on every change in the source. The job identifies which Instance ID is impacted by the change and updates it in the relevant tables of the Fabric DB. This solution is only relevant for [Business Entity over PostgreSQL](#).

https://support.k2view.com/Academy/articles/40_stream_sync/01_stream_sync_overview.html



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- **Simplifying Certificates settings** – Fabric now offers a seamless way to set certificates for interfaces directly within the Interface Editor UI. Instead of relying on scripts to upload certificates to the *truststore* - a task which is typically handled by DevOps or IT teams - this setting can now be managed effortlessly through the UI. This improvement is particularly significant for cloud deployments, where access to servers for running scripts is either limited or not available at all (like in SaaS deployment).

When handling several environments within a project, this capability simplifies the deployment process further, as Fabric is auto-associates the certificate to the correct environment's interface, as configured in the environment interfaces UI.

- **Azure Blob Storage authentication methods** – Fabric now supports 2 additional authentication methods - Service Principal and Managed Identity.
- **Azure PG** has been introduced as a managed PostgreSQL. The configuration should be as shown below. To enable AWS IAM database authentication (when Fabric runs on AWS), leave the password field empty.

```
[system_db]
SYSTEM_DB_TYPE=postgresql
SYSTEM_DB_CONNECTION_STRING=jdbc:k2view:postgresql:aws://<HOST>:<PORT>/<DATABASE_NAME>
SYSTEM_DB_PASSWORD=<PASSWORD>
SYSTEM_DB_USER=<USERNAME>
```

- A new section **[jdbc_default_loader]** has been added to the config.ini file in order to configure JDBC loader and it is applicable for JDBC System DB (such as PostgreSQL or SQLite):

```
[jdbc_default_loader]
#MODE=BATCH
#QUEUE_SIZE=10000
#BATCH_SIZE=1000
```

- **Default location of SQLite files as System DB** is now set in the Fabric config.ini file as follows (until now, there was not default location and the user had to specify it):

```
${FABRIC_HOME}/storage/systemdb
```

- 2 new reserved words have been added to **AFFINITY**, impacting jobs and batches:
 - **local** – to use the current node
 - **random** – to use any one node from the pool
- **Support for Multiple Secret Manager Instances** – Fabric now supports multiple instances of the same secret manager type, which can be defined in the `config.ini` file. This feature is particularly useful in scenarios where different secret manager instances or services handle different environments, such as UAT and Production, while the same Fabric needs to connect to all. (*Related ticket: #40539*)



RESOLVED ISSUES

- Ticket #38961 – getting an error in the DATA_DISCOVERY_JOB when the Web studio is running on a Docker. The problem has been resolved.
- Ticket #39942 – Schema Editor UX improvements: When connecting columns between large tables, the relevant columns might not be immediately visible, making it difficult to understand the populations' connections. To address this, it is now possible to hover over a population link to view a tooltip. The tooltip displays details such as the source and destination tables, columns, and the population name for improved clarity.
- Ticket # 39651 – Fabric converting Datetime with Timestamp and failing in Load. The problem has been resolved.
- Ticket #39758 – Schema Editor UX improvements: the search for a table's column now offers a more seamless experience. Clicking a column in the search bar takes you to the column within its table. The column list automatically scrolls to the selected column, which is then highlighted for an easy identification.
- Ticket #39992 – Web Studio issue with having multiple <EntityId> table's property. The problem has been resolved.
- Ticket #40218 – Catalog masking actor cannot handle an array of values in a JSON field. The problem has been resolved.
- Ticket #40330 – JsonParser1 NumberFormatException. The problem has been resolved.
- Ticket #40414 – Discovery job fails for BigQuery interface. The problem has been resolved.
- Ticket #40787 – data explorer – fixing the display of population time in MS (milliseconds).
- Ticket # 40795 – Ownership Candidates Num column in a Job status sometimes shows incorrect number of nodes that can run the job (affinity). The problem has been resolved.
- Ticket #40869 – during a Fabric restart, System DB tables index creation should take place only if it does not exist.
- Ticket #40710 – Fabric studio - dragging DB Objects from the Cassandra interface to the schema creates a table with all TEXT columns. The problem has been resolved.
- Ticket #40131 – Exception details are not representing the issue. The problem has been resolved.
- Ticket #41022 – Change Java Compiler/Code Compliance Level to Java 21 on WebStudio. The problem has been resolved.